

$$\begin{aligned}
& + \sum_{AB} h_{AB}(\phi) \left[ - \left( \bar{\lambda}_A \not{D} (1 - \gamma_5) \lambda_B \right) - \frac{1}{2} f_{A\mu\nu} f_B^{\mu\nu} + \frac{i}{4} \epsilon_{\mu\nu\rho\sigma} f_A^{\mu\nu} f_B^{\rho\sigma} \right. \\
& \quad \left. + D_A D_B \right] \\
& + \frac{\sqrt{2}}{2} \sum_{ABn} \frac{\partial h_{AB}(\phi)}{\partial \phi_n} \left[ - \left( \bar{\psi}_B \gamma^\mu \gamma^5 \psi_{nL} \right) f_{A\mu\nu} + 2i \left( \bar{\psi}_B \psi_{nL} \right) D_A \right] .
\end{aligned}$$

The other terms in Eq. (27.4.41) are just given by the gauge-invariant version of the Lagrangian density (26.8.6). Putting this together gives the Lagrangian density

$$\begin{aligned}
\mathcal{L} = & \text{Re} \sum_{nm} \mathcal{G}_{nm}(\phi, \phi^*) \left[ - \frac{1}{2} \left( \bar{\psi}_m \not{D} (1 + \gamma_5) \psi_n \right) + \mathcal{F}_n \mathcal{F}_m^* - D_\mu \phi_n D^\mu \phi_m^* \right] \\
& - 2 \text{Re} \sum_i \frac{\partial K(\phi, \phi^*)}{\partial \phi_i^*} D_A (\phi^* t_A)_i \\
& + i\sqrt{2} \frac{\partial^2 K(\phi, \phi^*)}{\partial \phi_i \partial \phi_j^*} \left[ (t_A \phi)_i \bar{\psi}_j \lambda_{AR} - (\phi^* t_A)_j \bar{\psi}_i \lambda_{AL} \right] \\
& - \text{Re} \sum_{nml} \frac{\partial^3 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_l^*} \left( \bar{\psi}_n \psi_{mL} \right) \mathcal{F}_l^* \\
& + \text{Re} \sum_{nml} \frac{\partial^3 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_l^*} \left( \bar{\psi}_m \gamma^\mu \psi_{lR} \right) D_\mu \phi_n \\
& + \frac{1}{4} \sum_{nmkl} \frac{\partial^4 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_l^* \partial \phi_k^*} \left( \bar{\psi}_n \psi_{mL} \right) \left( \bar{\psi}_k \psi_{lR} \right) \\
& - \text{Re} \sum_{nm} \frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} \left( \bar{\psi}_n \psi_{mL} \right) + 2 \text{Re} \sum_n \mathcal{F}_n \frac{\partial f(\phi)}{\partial \phi_n} \\
& + \frac{1}{4} \text{Re} \sum_{ABnm} \left( \bar{\lambda}_A \lambda_{BL} \right) \left( \bar{\psi}_n \psi_{mL} \right) \frac{\partial^2 h_{AB}(\phi)}{\partial \phi_n \partial \phi_m} - \frac{1}{2} \text{Re} \sum_{ABn} \left( \bar{\lambda}_A \lambda_{BL} \right) \mathcal{F}_n \frac{\partial h_{AB}(\phi)}{\partial \phi_n} \\
& + \text{Re} \sum_{AB} h_{AB}(\phi) \left[ - \left( \bar{\lambda}_A \not{D} \lambda_{BR} \right) - \frac{1}{4} f_{A\mu\nu} f_B^{\mu\nu} + \frac{1}{8} i \epsilon_{\mu\nu\rho\sigma} f_A^{\mu\nu} f_B^{\rho\sigma} + \frac{1}{2} D_A D_B \right] \\
& + \frac{\sqrt{2}}{4} \text{Re} \sum_{ABn} \frac{\partial h_{AB}(\phi)}{\partial \phi_n} \left[ - \left( \bar{\lambda}_B \gamma^\mu \gamma^5 \psi_{nL} \right) f_{A\mu\nu} + 2i \left( \bar{\lambda}_B \psi_{nL} \right) D_A \right] .
\end{aligned} \tag{27.4.42}$$

One interesting feature of this result is the appearance of a gaugino mass in theories with  $\phi_n$ -dependent functions  $h_{AB}(\phi)$ , when supersymmetry is broken by a non-vanishing value of  $\mathcal{F}_n$ . This mechanism is used to